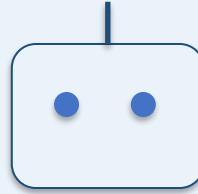
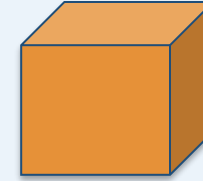


Warehouse



Robot



Inventory

CSA1014 SOFTWARE ENGINEERING

CO2 AT1 Warehouse Robotics Coordination and Inventory Automation Platform

**PRESENTED BY
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1. Scenario & Problem Statement

Scenario

A customer places an urgent order for 20 smartphones.

The warehouse checks stock, reserves products, assigns an available robot, completes picking, and updates inventory.

Problem

Manual coordination can cause:

- Stock errors
- Picking delays
- Poor robot utilization
- Incorrect inventory updates

The platform coordinates these activities.

2. Functional & Non-Functional Requirements

Functional Requirements

- Manage inventory
- Check stock
- Reserve products
- Assign robot tasks
- Track robot status
- Update inventory
- Complete tasks

Non-Functional Requirements

- Reliability – accurate updates
- Performance – quick processing
- Usability – simple manager interface
- Scalability – support more robots/products
- Security – protect warehouse data

3. Actors and Use Cases

Warehouse Manager

- Manage Inventory
- Check Stock
- Assign Robot Task
- Track Robot

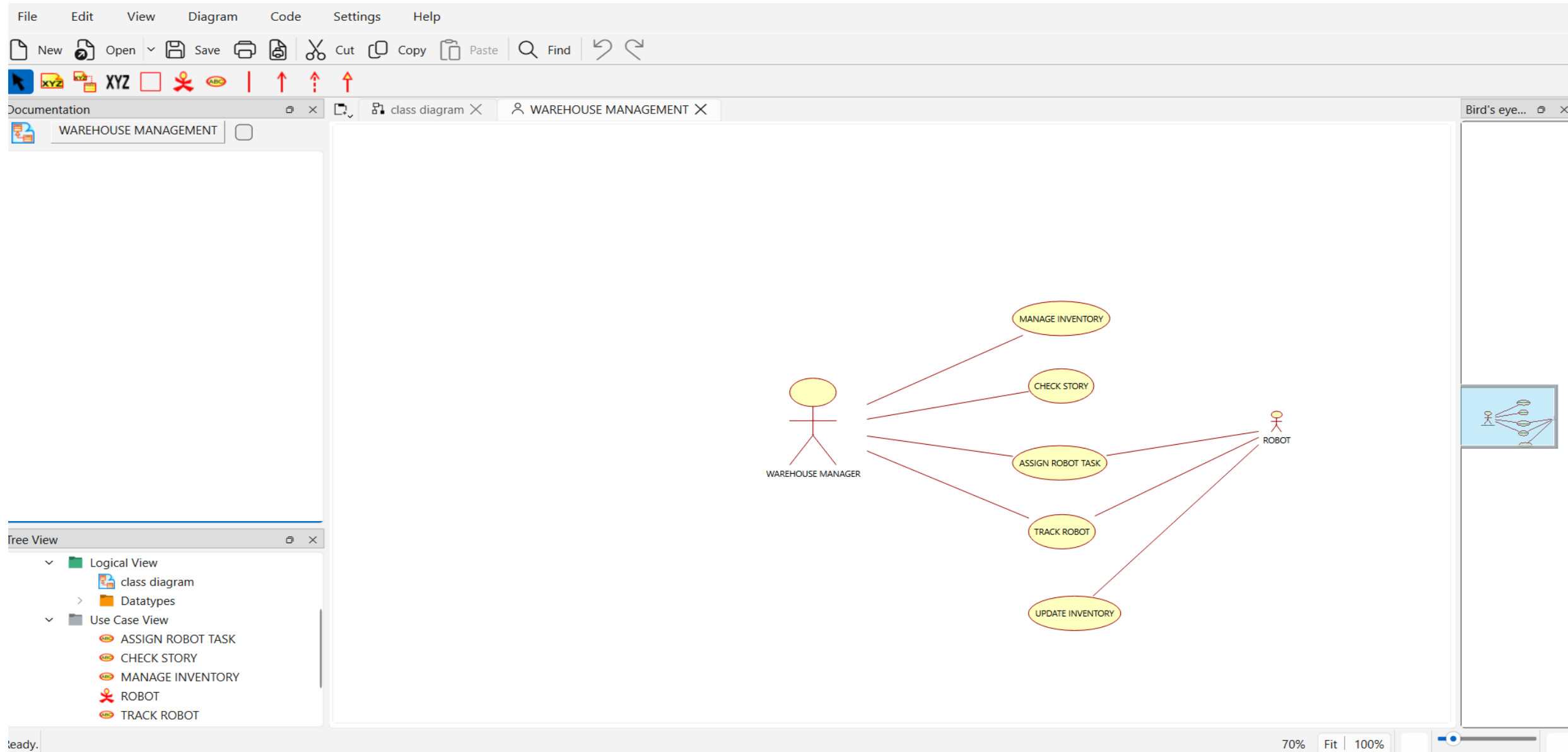
Robot

- Receive Task
- Pick Product
- Update Status
- Support Inventory Update

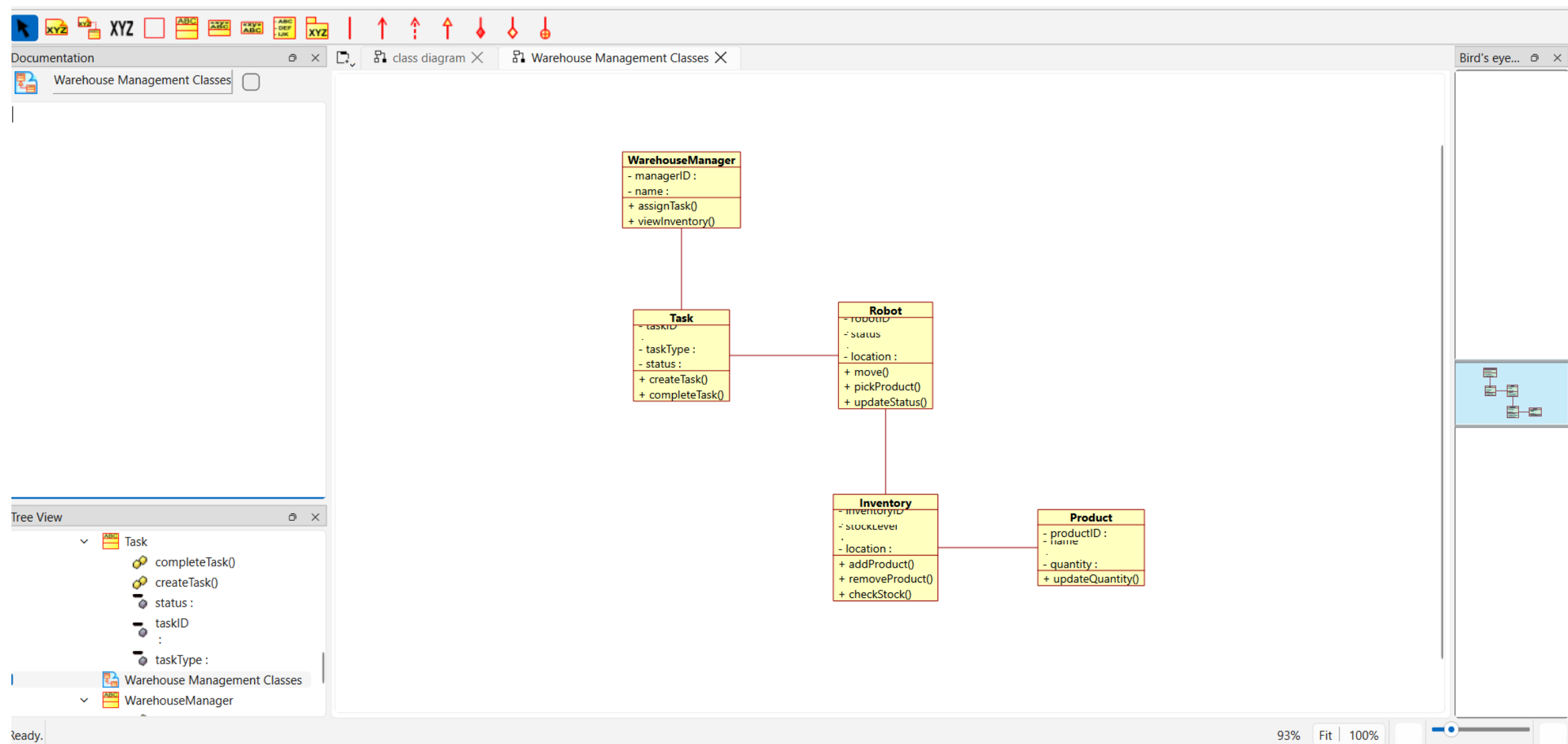
Main Use Cases

Manage Inventory
Check Stock
Assign Robot Task
Track Robot
Update Inventory

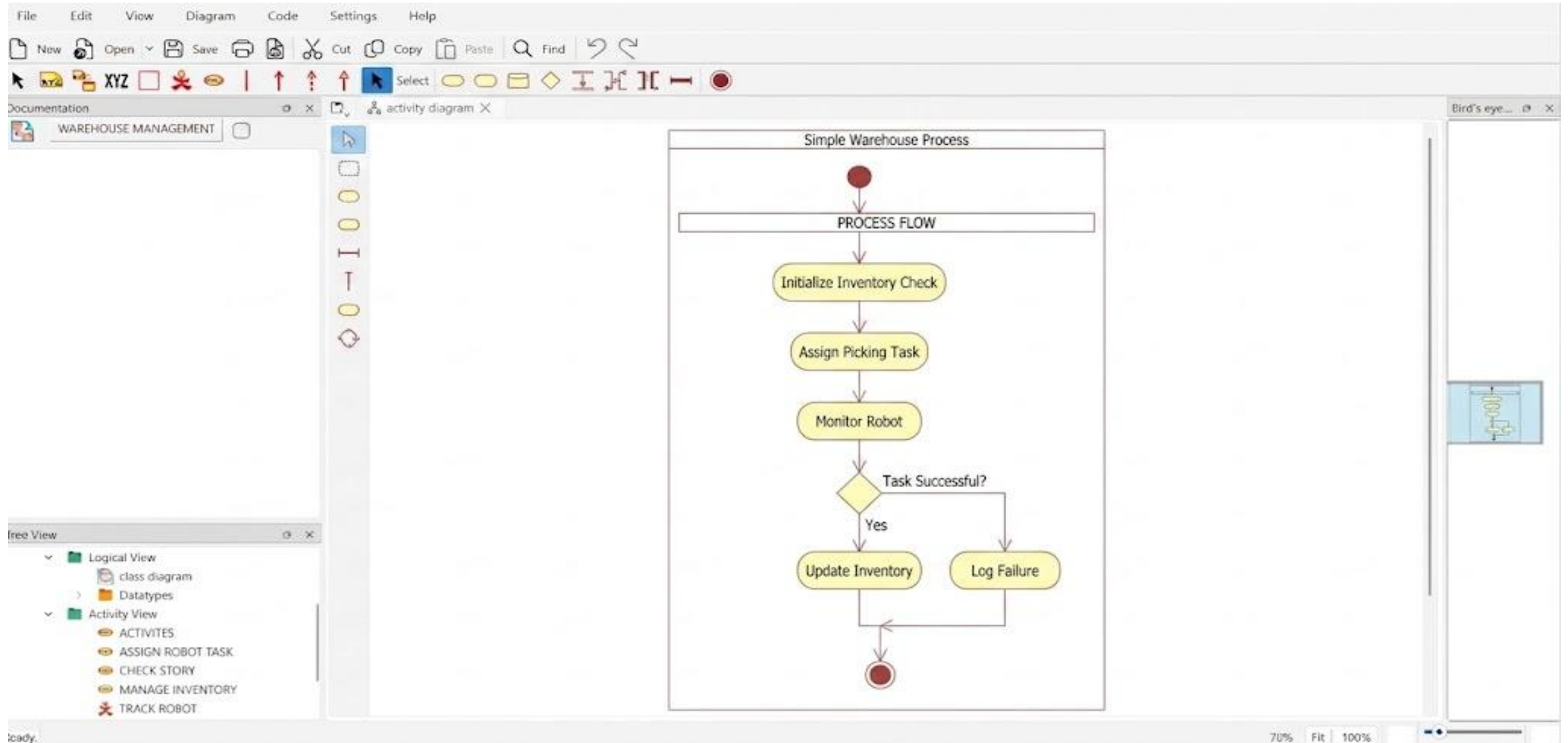
4. Use Case Diagram



5. Class Diagram – Main Classes



6. Activity Diagram



7. Scenario Workflow



Example: 20 smartphones are ordered → stock is verified → robot is assigned → products are picked → inventory is updated.

8. Scenario-to-UML Mapping

Scenario Activity

Customer order



Check stock



Reserve products



Assign robot



Pick products



Update inventory

UML Elements

Actors:

- Warehouse Manager
- Robot

Use Cases:

- Check Stock
- Assign Robot Task
- Track Robot
- Update Inventory

Classes

Warehouse Manager
Task
Robot
Inventory
Product

These classes support the scenario.

9. Expected Outcome / Result

- A clear UML model of the warehouse automation platform is produced in Umbrello.
- The Use Case Diagram shows actors and system functions.
- The Class Diagram shows the main entities and relationships.
- The scenario connects order processing, robot operations and inventory updates.
- The model can guide future software development.

10. Conclusion

Conclusion

Umbrello provides a simple way to model the Warehouse Robotics Coordination and Inventory Automation Platform before coding.

The scenario-based design connects warehouse actors, robot operations, inventory management and task processing into one understandable system model.

Thank You